

Electrostatics Lecture 1

Electric Forces and Fields

Chapter 18

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Physics 203 PCC

Electric Charge

What happens when you rub a balloon on your hair?

Electric Charge

The balloon has gained an invisible property that allows it to pull on other objects. But what is it? We can give it a name: *electric charge*. By understanding it, we might answer:

- Why do some objects attract while others repel?
- What does it mean for something to be “neutral”?
- How does charge move from one object to another?



Electrostatics is the study of electric charge at rest and the forces they exert on one another.

Electric Charge

Electric charge is a fundamental unit of matter that determines how objects interact through electrostatic forces.

Point Charge (q)

- Charge of an individual particle (microscopic)
- Quantized in units of the elementary charge

$$q = ne$$

- $e = 1.602 \times 10^{-19} \text{ C}$

System-Level Charge (Q)

- Total charge of an object or system (macroscopic)
- Sum of all individual charges

$$Q = \sum_i q_i$$

Conductors and Insulators

Materials can be classified into two broad categories:

Conductors

- Materials that allow electric charge to move freely.
- Examples: Metals, Saltwater, Plasma
- At electrostatic equilibrium, excess charge lies on the surface, and the electric field inside the conductor is zero.

Insulators

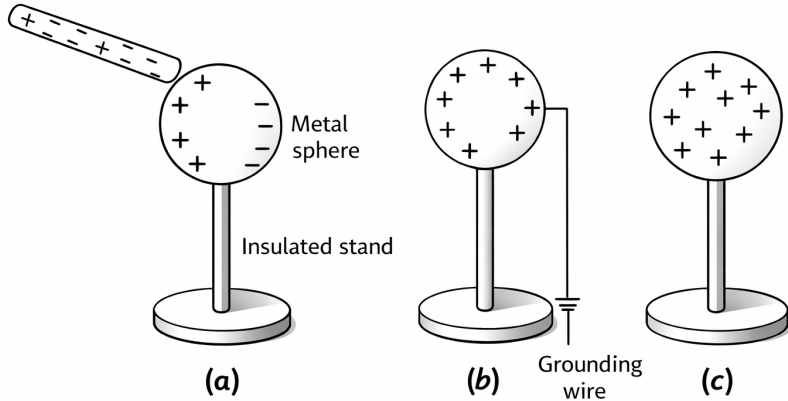
- Materials that do not allow charge to move freely.
- Examples: Rubber, Plastic, Glass, Ceramics, Pure (distilled) water.
- Excess charge stays localized, and electric fields can exist within the material.

Charging by Conduction

Charging by contact occurs when a charged object physically touches a neutral object, transferring charge between them. The neutral object ends up with the **same type of charge** as the charged object. The process:

1. A charged object is brought into direct contact with a neutral conductor.
2. Electrons transfer between the objects until they reach electrostatic equilibrium.
3. After separation, the previously neutral object is left with the same type of charge as the charged object.

Charging by Induction



Conservation of Charge

One of the fundamental principles of physics is the **law of charge conservation**:

- The total electric charge of an isolated system remains constant.
- Charge may be transferred, but it cannot be created or destroyed.

$$Q_{\text{final}} = Q_{\text{initial}}$$

This law applies to all physical processes, including:

- Conduction and induction
- Chemical reactions (e.g., batteries)
- Electric circuits

Consequence: In conductors, charge redistributes itself while preserving the total charge.

Coulomb's Law

The electrostatic force F between two point charges q_1 and q_2 separated by a distance r is given by:

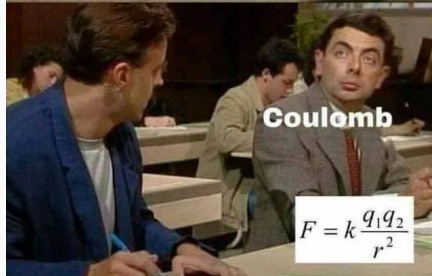
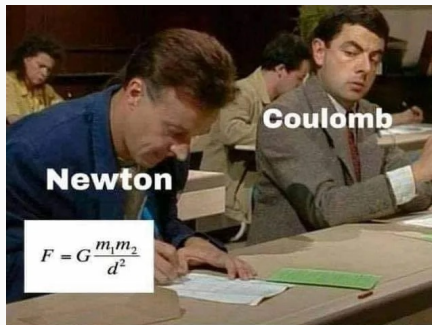
$$F = k \frac{|q_1 q_2|}{r^2} \quad (1)$$

where:

- F is the magnitude of the electrostatic force (in newtons, N),
- q_1 and q_2 are the magnitudes of the two charges (in coulombs, C),
- r is the distance between the charges (in meters, m),
- $k = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$ is Coulomb's constant.

If q_1 and q_2 have the same sign (both positive or both negative), the force is **repulsive**. If q_1 and q_2 have opposite signs, the force is **attractive**. - The force acts along the straight line joining the two charges.

Wait a second...



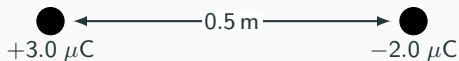
The form of Coulomb's law should look familiar to you,

- $F = k \frac{|q_1 q_2|}{r^2}$
- $F = G \frac{m_1 m_2}{r^2}$

Both are inverse square laws!

Example 1: 2 Point Charges in a Vacuum

Two point charges are placed in vacuum. Charge $q_1 = +3.0 \mu\text{C}$ is positioned at $x = 0$, and charge $q_2 = -2.0 \mu\text{C}$ is placed at $x = 0.5 \text{ m}$. Determine the magnitude and direction of the electrostatic force exerted by q_1 on q_2 .



Solution

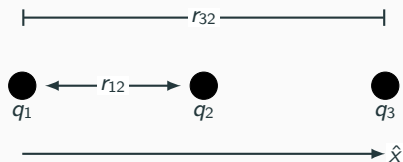
$$F = k \frac{|q_1 q_2|}{r^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \frac{(3.0 \times 10^{-6} \text{ C})(2.0 \times 10^{-6} \text{ C})}{(0.5 \text{ m})^2}$$
$$= 0.216 \text{ N}$$

$q_1 > 0$ and $q_2 < 0 \rightarrow$ Force between them is attractive.

q_1 pulls q_2 to the left, q_2 pulls q_1 to the right

Example 2: 3 Charges in a Line

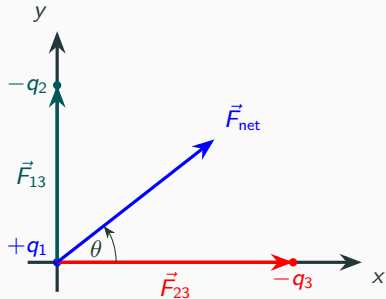
Consider three point charges arranged along a straight line.



The net force on q_2 is the sum of forces due to q_1 and q_3 . Each force is calculated using Coulomb's Law and then summed as vectors. For example, if $q_1 < 0$, $q_2 > 0$, and $q_3 > 0$,

$$\vec{F}_{\text{net},2} = \vec{F}_{12} + \vec{F}_{32} = -k \frac{|q_1 q_2|}{r_{12}^2} \hat{x} - k \frac{|q_3 q_2|}{r_{32}^2} \hat{x} = -k \left(\frac{|q_1 q_2|}{r_{12}^2} + \frac{|q_3 q_2|}{r_{32}^2} \right) \hat{x}$$

Vector Components in 2D



Just like any other force, Coulomb forces are vectors! Using Coulomb's Law, each force is:

$$\vec{F}_{12} = k \frac{|qq_2|}{r_2^2} \hat{y}, \quad \vec{F}_{13} = k \frac{|qq_3|}{r_3^2} \hat{x}$$

Notice that both q_2 and q_3 are negative, q_1 is positive. Then we know,

$$\vec{F}_{net} = k \frac{|qq_3|}{r_3^2} \hat{x} + k \frac{|qq_2|}{r_2^2} \hat{y}$$

Written another way, its magnitude,

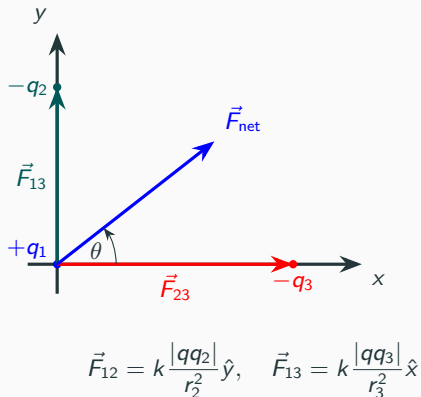
$$\begin{aligned} F_{net} &= \sqrt{F_{net,x}^2 + F_{net,y}^2} \\ &= \sqrt{F_{13}^2 + F_{12}^2} \end{aligned}$$

At an angle,

$$\begin{aligned} \theta &= \tan^{-1} \left(\frac{F_{net,x}}{F_{net,y}} \right) \\ &= \tan^{-1} \left(\frac{F_{13}}{F_{12}} \right) \end{aligned}$$

Example: Vector Components in 2D

If $q_1 = 3\mu\text{C}$, $q_2 = q_3 = -3\mu\text{C}$ and $r_2 = r_3 = 0.5\text{ m}$, what is F_{net} and θ ?



$$\vec{F}_{\text{net}} = k \frac{|qq_3|}{r_3^2} \hat{x} + k \frac{|qq_2|}{r_2^2} \hat{y}$$

$$F_{\text{net}} = \sqrt{F_{\text{net},x}^2 + F_{\text{net},y}^2}$$
$$= \sqrt{F_{13}^2 + F_{12}^2}$$

At an angle,

$$\theta = \tan^{-1} \left(\frac{F_{\text{net},x}}{F_{\text{net},y}} \right)$$
$$= \tan^{-1} \left(\frac{F_{13}}{F_{12}} \right)$$

What is a Field?

- A field is a way to describe how a force exists in space, even before anything experiences it.
- Think of it as an invisible influence that fills a region and can exert a force on objects placed within it.
- A field allows us to think about how one object creates a “presence” that extends outward, affecting other objects that come near.
- Gravitational Field: Imagine some mass M , think of a point mass m like a probe, the gravitational field tells us what the gravitational force would be wherever in space we put m near M .
- Electric Field: Likewise, imagine we have some charge Q , a test charge q like a probe, the Electric Field tells us what the electric force will be wherever in space we put q near Q .

The Electric Field in terms of Force

The electric field, denoted as $\vec{E}$, is defined as the force per unit charge:

$$\vec{E} = \frac{\vec{F}}{q}$$

where:

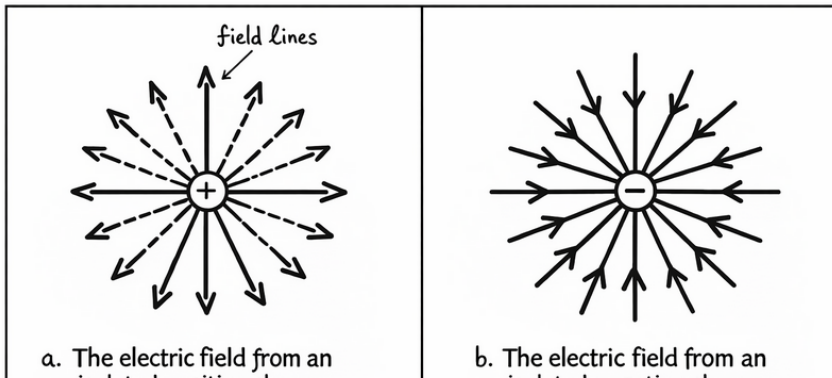
- $\vec{E}$ is the electric field (in newtons per coulomb, N/C),
- $\vec{F}$ is the electric force experienced by a test charge,
- q is the charge experiencing the force.

This definition means that if we place a small positive charge (*test charge*) in an electric field, the force it experiences tells us the strength and direction of the field at that location.

The Direction of an Electric Field

The direction of the electric field at a point is the direction of the force that would act on a small **positive** test charge placed at that point:

- If the field is created by a **positive** charge, the electric field points outward (away from the charge).
- If the field is created by a **negative** charge, the electric field points inward (toward the charge).



The Electric Field on a Point Charge

The magnitude of the electric field created by a single point charge is given by:

$$E = k_e \frac{|Q|}{r^2}$$

where:

- $k_e = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ is Coulomb's constant,
- Q is the charge creating the field,
- r is the distance from the charge to the point of interest.

This equation shows that the electric field gets stronger as you get closer to the charge and weaker as you move farther away.

Example 5: The Electric Field of a Point Charge

There is an isolated point charge of $q = +15\ \mu\text{C}$ in a vacuum. Using a test charge of $q_0 = +0.80\ \mu\text{C}$, determine the electric field at point P , which is $0.20\ \text{m}$ away to the right at point P .



Solution:

The electric field anywhere due to $+q$ is,

$$\vec{E} = k \frac{|q|}{r} \hat{r}$$

At point P , $\hat{r} = \hat{x}$ so,

$$\vec{E} = k \frac{|q|}{r} \hat{x} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{15 \times 10^6 \text{ C}}{(0.20 \text{ m})^2} = 3.4 \times 10^4 \text{ N/C}$$

What is the force acting on a particle with charge $0.50\ \mu\text{C}$ at point P ?

Bonus Round

Which way would the electric field point if q was negative?

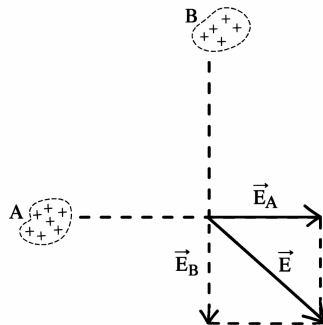


The Superposition of Electric Fields

At a particular point in space, each of the surrounding charges contributes to the net electric field that exists there. To determine the net field, it is necessary to obtain the various contributions separately and then find the vector sum of them all. Such an approach is an illustration of the principle of linear superposition, as applied to electric fields. The following example demonstrates this.

Example 4: The Superposition of Electric Fields

Consider two charged objects, A and B. Each contributes as follows to the net electric field at point P at the coordinates (0,0). $\vec{E}_A = 3.00\text{ N/C}$ directed to the right, and $\vec{E}_B = 2.00\text{ N/C}$ directed downward. Thus, $\vec{E}_A$ and $\vec{E}_B$ are perpendicular. What is the net electric field at P?



The magnitude of the net electric field is

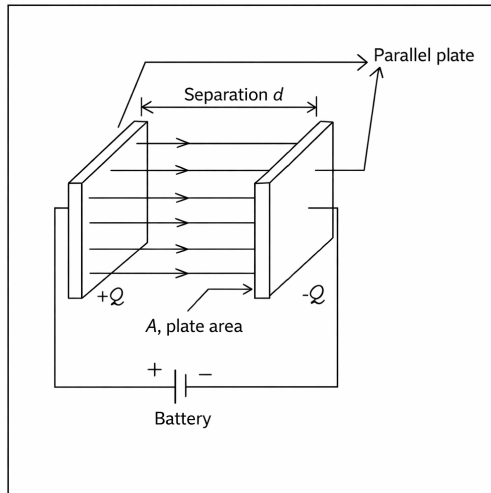
$$\begin{aligned} E &= \sqrt{E_A^2 + E_B^2} \\ &= \sqrt{(3.00\text{ N/C})^2 + (2.00\text{ N/C})^2} \\ &= \boxed{3.61\text{ N/C}} \end{aligned}$$

The direction of $\vec{E}$ is given by the angle θ in the drawing:

$$\theta = \tan^{-1} \left(\frac{E_B}{E_A} \right)$$

The Parallel Plate Capacitor

A **capacitor** is a device used to store electric charge. The simplest type of capacitor is a **parallel plate capacitor**, which consists of two flat conducting plates placed near each other, separated by air or another insulating material

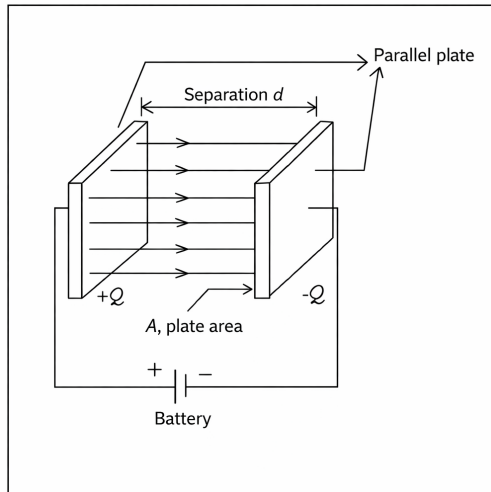


The Parallel Plate Capacitor

When charge is placed on a parallel plate capacitor, one plate becomes positively charged, and the other becomes negatively charged by the same amount. This creates a separation of charge, which results in an electric field between the plates.

The amount of charge a capacitor can store depends on:

- The **size of the plates** – larger plates can hold more charge.
- The **distance between the plates** – closer plates create a stronger attraction between opposite charges.



- The **material between the plates** –

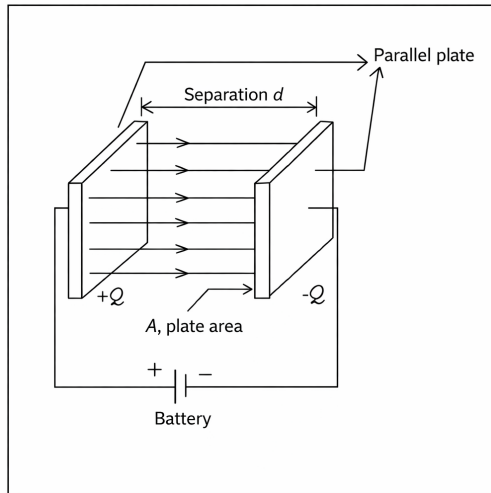
The Parallel Plate Capacitor

An **electric field** forms in the space between the plates.

$$E = \frac{q}{\epsilon_0 A} = \frac{\sigma}{\epsilon_0}$$

where:

- q is the charge on one of the plates,
- A is the area of each plate,
- $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$ is the permittivity of free space,
- $\sigma = \frac{q}{A}$ is the **surface charge density**, which represents charge per unit area on the plates.



This equation tells us that the electric field depends only on the charge per unit area and not on

Electric Field Lines

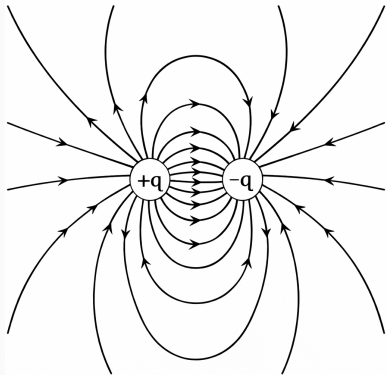


Figure 2: The electric field lines of an electric dipole are curved and extend from the positive to the negative charge. At any point, such as 1, 2, or 3, the field created by the dipole is tangent to the line through the

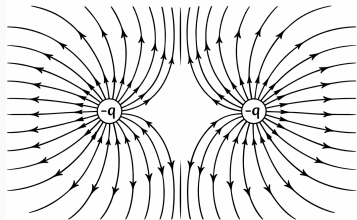


Figure 3: The electric field lines for two identical positive point charges. If the charges were both negative, the directions of the lines would be reversed.

The Electric Field Inside a Conductor

When a conductor is placed in an external electric field, free electrons inside the conductor move until the charges redistribute themselves in such a way that they cancel out any electric field within the material of the conductor.

- Any **net charge** on a conductor resides on its **outer surface**.
- The **electric field inside the conductor is zero**.
- The conductor acts as a barrier, preventing external fields from influencing the space inside it.

This is known as **electrostatic shielding**. Example: A **Faraday cage** is a conducting enclosure that blocks external electric fields.

- **Car bodies**: If a lightning bolt strikes a car, the electric field inside remains zero, protecting passengers.
- **Microwave ovens**: The metal grid in the door prevents microwaves from escaping.
- **Shielded cables**: Coaxial cables use a conducting outer layer to prevent interference.

Gauss' Law

A charge distribution is a collection of charges that are spread out over a region. Gauss' law describes the relationship between a charge distribution and the electric field it produces.

To understand Gauss' Law, we first introduce the idea of **electric flux**, which measures how much an electric field passes through a surface. Flux depends on:

- The **strength** of the electric field E .
- The **area** of the surface through which the field passes.
- The **direction** of the field relative to the surface.

Gauss' Law

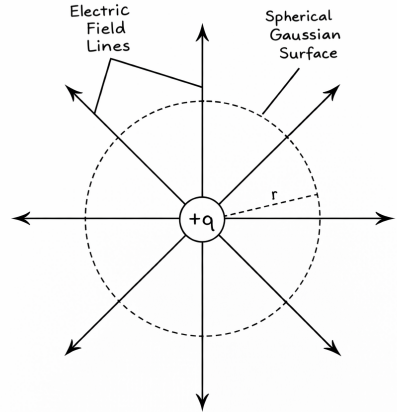
Consider a point charge. The electric field it produces is given by

$$E = q/(4\pi\epsilon_0)r^2$$

Imagine an imaginary surface around the charge, the electric field given above is the field everywhere on that surface – called a **Gaussian Surface**.

We know that the area of a sphere is just $A = 4\pi r^2$, so **Gauss' Law for a point charge is**

$$E = \frac{q}{A\epsilon_0} \rightarrow EA = \frac{q}{\epsilon_0}$$



Gauss' Law

$$EA = \frac{q}{\epsilon_0}$$

EA is known as the **electric flux**, and is represented as Φ_E and its a measure of the total electric field lines passing through a given surface. Rewriting Gauss' law more generally,

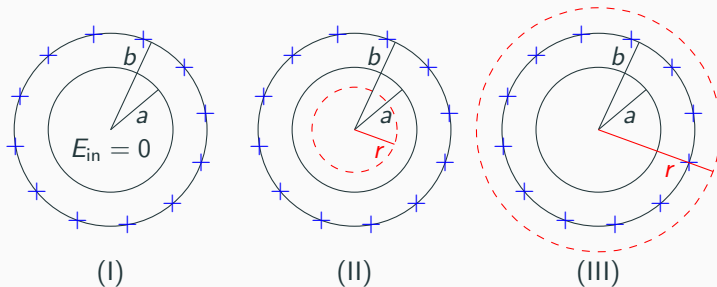
$$\Phi_E = \frac{Q_{enc}}{\epsilon_0}.$$

The electric flux through any closed surface is equal to the net charge inside the surface, Q_{inside} , divided by ϵ_0 .

Holds for any any distribution of charges within an arbitrary shaped Gaussian surface!

Example 6: The Electric Field of a Charged Spherical Shell

A spherical conducting shell of inner radius a and outer radius b carries a total charge $+Q$ distributed on the surface of a conducting shell. Find the electric field (a) in the interior of the conducting shell for $r < a$ and (b). outside the shell for $r > b$. (c) If an additional charge of $-2Q$ is placed at the center, find the electric field for $r > b$. (d). What is the distribution of charge on the sphere in part (c)?



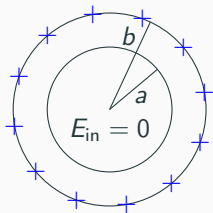
Example 6: The Electric Field of a Charged Spherical Shell

Solution:

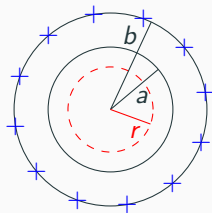
(a). (Case II) $Q_{enc} = 0 \rightarrow E = 0$

(b). (Case III) $EA = E(4\pi r^2) = \frac{Q_{enc}}{\epsilon_0} = \frac{Q}{\epsilon_0} \rightarrow E = \frac{Q}{4\pi\epsilon_0 r^2}$

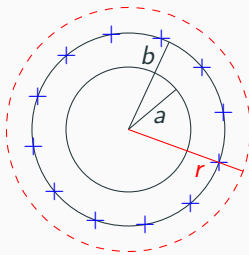
(c). $EA = E(4\pi r^2) = \frac{Q_{inside}}{\epsilon_0} = \frac{Q-2Q}{\epsilon_0} \rightarrow E = -\frac{Q}{4\pi\epsilon_0 r^2}$



(I)



(II)



(III)

Example 6: The Electric Field of a Charged Spherical Shell

Solution:

(d). Find the charge distribution on the sphere for part (c).

$$EA = E(4\pi r^2) = \frac{Q_{inside}}{\epsilon_0} = \frac{Q_{center} + Q_{innersurface}}{\epsilon_0}$$

The charge on the inner surface of the shell:

$$Q_{center} + Q_{innersurface} = 0$$

because the electric field in the conductor is zero. So

$$Q_{innersurface} = -Q_{center} = 2Q$$

Find the charge on the outer surface, noting that the inner and outer surface charges must sum to $+Q$:

$$Q_{outersurface} + Q_{innersurface} = Q$$

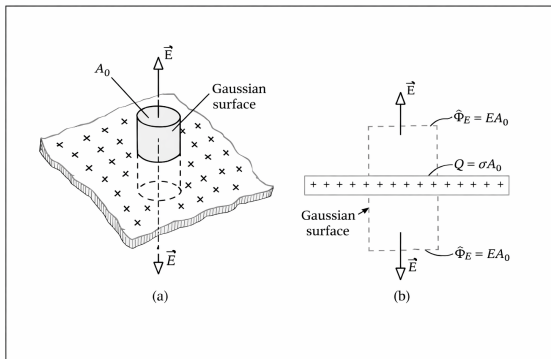
$$Q_{outersurface} = -Q_{innersurface} + Q = -Q$$

Example 7: A Nonconducting Plane Sheet of Charge

Find the electric field above and below a nonconducting infinite plane sheet of charge with uniform positive charge per unit area σ

Reasoning:

By symmetry the electric field must be perpendicular to the plane and directed away from it on either side as shown in (b). For the Gaussian surface, choose a small cylinder with axis perpendicular to the plane, each end having an area of A_0 .



No electric field lines pass through the curved surface of the cylinder, only through the two ends, which have total area $2A_0$.

Example 7: A Nonconducting Plane Sheet of Charge

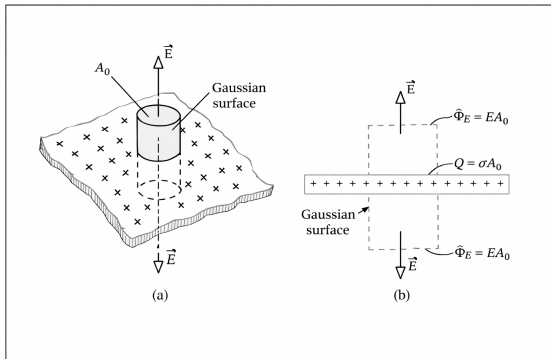
Find the electric field above and below a nonconducting infinite plane sheet of charge with uniform positive charge per unit area σ

Solution: Find the electric field above and below a plane of uniform charge. Using Gauss's law:

$$EA = \frac{Q_{\text{inside}}}{\epsilon_o}$$

The total charge inside the Gaussian cylinder is the charge density times the cross-sectional area:

$$Q_{\text{inside}} = \sigma A_o$$



The electric flux comes entirely from the two ends, each having area A_o . Substitute $A = 2A_o$ and Q_{inside} and solve for E .

$$E = \frac{\sigma A_o}{(2A_o)\epsilon_o} = \frac{\sigma}{2\epsilon_o}$$

Example 7: A Nonconducting Plane Sheet of Charge

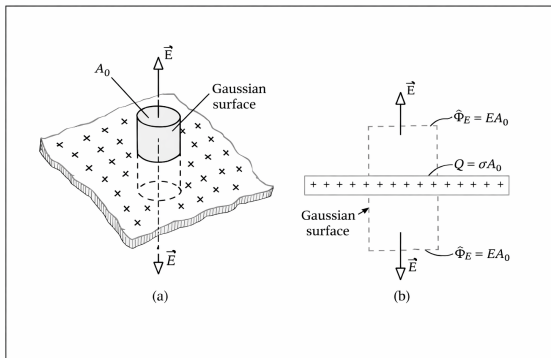
Find the electric field above and below a nonconducting infinite plane sheet of charge with uniform positive charge per unit area σ

Solution: (cont.)

The electric field points away from the plane, so it's positive above the plane and negative below the plane.

$$E_z = \frac{\sigma}{2\epsilon_0}, z > 0$$

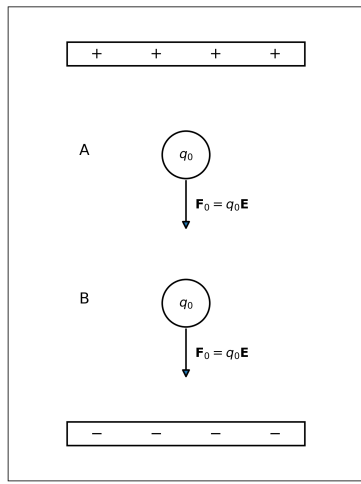
$$E_z = -\frac{\sigma}{2\epsilon_0}, z < 0$$



Electric Potential and Potential Energy

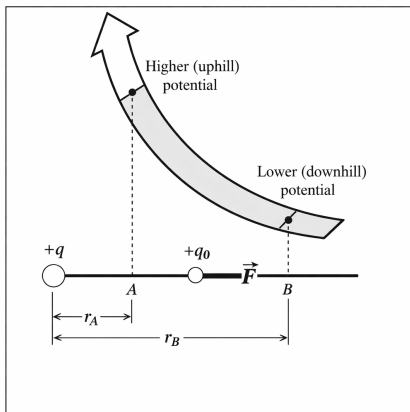
Gravitational System	Electric System
Mass m Gravitational Field g Potential Energy: $U_g = mgh$ Potential: gh Force: $F = mg$	Charge q Electric Field E Potential Energy: $U_e = qV$ Potential: V Force: $F = qE$

Electric Potential Difference and Work



Because of the electric field $\vec{E}$, an electric force $\vec{F} = q_0 \vec{E}$ is exerted on a positive test charge

The Electric Potential Difference Created by a Point Charge



A positive test charge $+q_0$ experiences a repulsive force from another positive charge $+q$, moving from point A to point B under the influence of the electric field.

- For a point charge, the electric potential at a distance r from q :

$$V = \frac{kq}{r}$$

- Potential difference definition:

$$\begin{aligned}\Delta V = V_B - V_A &= \frac{kq}{r_B} - \frac{kq}{r_A} \\ &= -\frac{W_{AB}}{q_0}\end{aligned}$$

Example: The Potential Energy of a Group of Charges

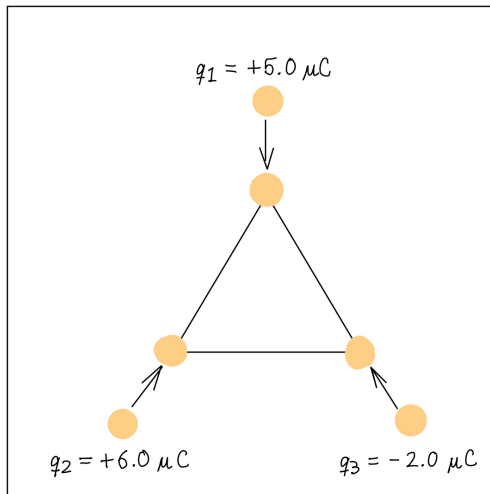


Figure 4: Three point charges are initially located infinitely far apart. They are then brought together and placed at the corners of an equilateral triangle. Each side of the triangle has a

Equipotential Surfaces

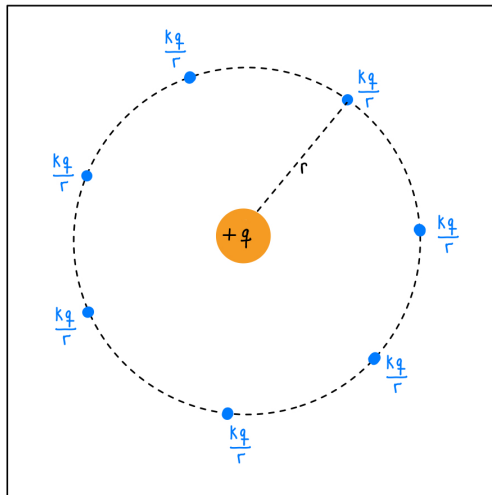
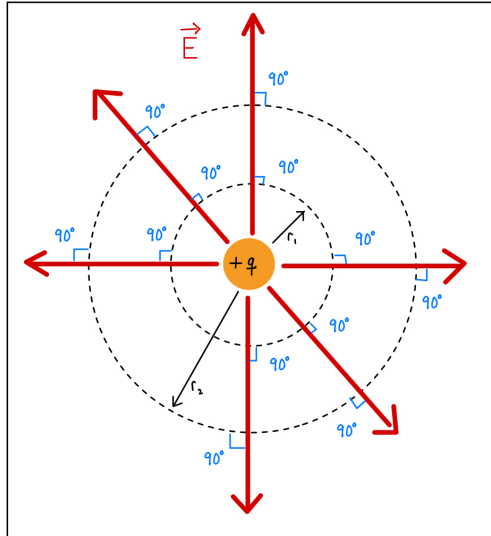


Figure 5: A cross-sectional diagram of a sphere with a point charge inside showing that at a

Equipotential Surfaces



Parallel Plate Capacitor

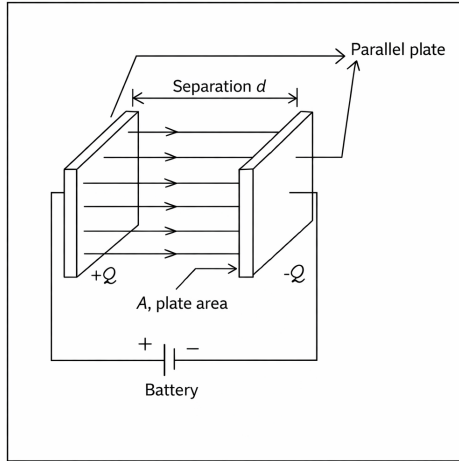


Figure 7: A basic parallel plate capacitor with charge $+Q$ and $-Q$ on each plate and a voltage ΔV across them.

Dielectrics

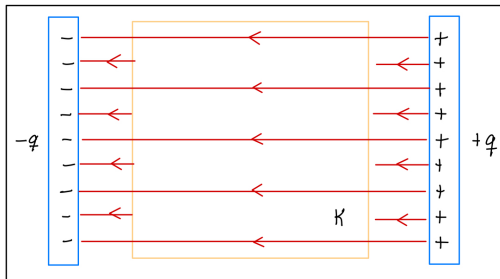


Figure 8: The surface charges on the dielectric reduce the electric field inside the dielectric.

$$C = \kappa \epsilon_0 \frac{A}{d}$$

Dielectrics Constants

Substance	Dielectric Constant, κ
Vacuum	1
Air	1.000 54
Teflon	2.1
Benzene	2.28
Paper (royal gray)	3.3
Ruby mica	5.4
Neoprene rubber	6.7
Methyl alcohol	33.6
Water	80.4

Table 1: Dielectric constants κ for various substances.

Energy Stored in a Capacitor

Stored energy (as work done by the battery):

$$U = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{1}{2}\frac{Q^2}{C}$$

In terms of electric field (parallel-plate capacitor):

$$V = Ed, \quad C = \kappa\epsilon_0 \frac{A}{d}$$

$$U = \frac{1}{2}\kappa\epsilon_0 AdE^2$$

Energy density (per unit volume):

$$\text{EnergyDensity} = \frac{U}{\text{Volume}} = \frac{1}{2}\kappa\epsilon_0 E^2$$

V refers to potential difference between plates. Stronger *E* → more energy per unit volume.

Summary of Key Electrostatic Equations

Concept	Equation
Coulomb's Law (force between point charges)	$F = k \frac{ q_1 q_2 }{r^2}$
Electric Field from a Point Charge	$E = k \frac{ q }{r^2}$
Electric Force on a Charge in a Field	$\vec{F} = q\vec{E}$
Electric Potential from a Point Charge	$V = k \frac{q}{r}$
Potential Energy of a Charge in a Field	$U_e = qV$
Electric Potential Difference	$\Delta V = \frac{W}{q}$
Electric Field and Potential (uniform field)	$E = \frac{\Delta V}{d}$
Capacitance Definition	$C = \frac{Q}{\Delta V}$
Capacitance of a Parallel Plate Capacitor	$C = \kappa \epsilon_0 \frac{A}{d}$
Energy Stored in a Capacitor	$U = \frac{1}{2} QV = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$
Energy Density in an Electric Field	$u = \frac{1}{2} \kappa \epsilon_0 E^2$